

Aegis Home System — Draft v1 V2.0

Minimum Viable Dielectric Citadel: Core Kitchen Appliances

Version 2.0 — June 2026

This is Iteration 1 of the Aegis Home system — the minimum viable product set establishing the “Ancient Hearth” kitchen as the core survival node.

Δ Change Log (V1.0 ⇄ V2.0)

- All ceramics: Flash Sintering protocol (1800°C, ±300°C vs previous)
- PEX-a plumbing: ASTM F876-25 updated standard compliance
- Added QD-YInMn Blue +31% UV enhancement to all exterior surfaces

Core V1 Components

1. **ZrB₂/Si₃N₄ Stove Burner (CSMFAB00000000000002)** — ancient hearth burner, Voronoi tessellation top, gel-cast fabrication, piezoelectric ignition, thermochromic Cobalt Flare indicators
2. **Aerogel Thermal Break (CSMFAB00000000000003)** — $\lambda=0.010$ W/m·K polyimide-silica hybrid, decouples burner heat from chassis
3. **PEX-a Plumbing System (CSMFAB00000000000012)** — $\rho > 10^{12}$ Ω·m, ASTM F876-25 compliant, eliminates copper GIC pathway
4. **BFRP Rebar (CSMFAB00000000000009)** — structural reinforcement, zero conductivity, 1000 MPa tensile per ACI 440.11-25
5. **YInMn Blue exterior coating** — SRI=115, reflects 85% NIR, primary spectral defense

Version 1 Limitation

Covers kitchen and plumbing only. No refrigeration, no microwave protection, no building envelope shielding, no wiring hardening. V2 onwards adds these layers.

V1 GIC threat reduction vs. standard home: ~35% (kitchen appliances + plumbing isolated)